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JEE Main Physics Practice Sheet — DPP-051

Topic: Moment of a Force; Torque — Definition, Cross Product & Calculation**Time Allowed:** 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** A force $\vec{F} = (2\hat{i} - 3\hat{j} + 4\hat{k})$ N acts at a point whose position vector is $\vec{r} = (3\hat{i} + 2\hat{j} + 3\hat{k})$ m. The torque $\vec{\tau} = \vec{r} \times \vec{F}$ about the origin is:
- (A) $(17\hat{i} - 6\hat{j} - 13\hat{k})$ N · m
 (B) $(17\hat{i} + 6\hat{j} - 13\hat{k})$ N · m
 (C) $(-17\hat{i} + 6\hat{j} + 13\hat{k})$ N · m
 (D) $(13\hat{i} - 6\hat{j} - 17\hat{k})$ N · m
- Q2.** A particle of mass m is projected with speed u at an angle θ to the horizontal from the origin $(0, 0)$. The magnitude of torque of the gravitational force on the projectile about the launch point when it reaches its maximum height is:
- (A) $mu^2 \sin \theta$
 (B) $\frac{mu^2 \sin 2\theta}{4}$
 (C) $\frac{mu^2 \sin 2\theta}{2}$
 (D) $\frac{mu^2 \sin^2 \theta}{2}$
- Q3.** The torque $\vec{\tau}$ acting on a body about a reference point O is always:
- (A) Parallel to both $\vec{r}$ and $\vec{F}$
 (B) In the plane containing $\vec{r}$ and $\vec{F}$
 (C) Perpendicular to both $\vec{r}$ and $\vec{F}$
 (D) In the direction of $\vec{F}$
- Q4.** A uniform rod AB of length $L = 2$ m is hinged at end A . A force $F = 50$ N acts at end B making an angle of 30° with the rod. The magnitude of the torque exerted by this force about the hinge A is:
- (A) $50\sqrt{3}$ N · m
 (B) 50 N · m
 (C) 100 N · m
 (D) 25 N · m
- Q5.** A particle moves in the xy -plane along the straight line $y = 4$ m with a constant velocity $\vec{v} = (5\hat{i})$ m/s. If the mass of the particle is $m = 2$ kg and a constant force $\vec{F} = (6\hat{i})$ N acts on it, the torque of the force about the origin is:
- (A) $+24\hat{k}$ N · m
 (B) $-24\hat{k}$ N · m
 (C) $-12\hat{k}$ N · m
 (D) 0
- Q6.** A force $\vec{F} = (a\hat{i} + 3\hat{j} + 6\hat{k})$ N acts at a point $\vec{r} = (2\hat{i} - 6\hat{j} - 12\hat{k})$ m. The value of constant a for which the torque about the origin is strictly zero is:
- (A) 1
 (B) -2
 (C) 2
 (D) -1
- Q7.** Two equal and opposite parallel forces of magnitude $F = 20$ N act on the rim of a circular steering wheel of radius $R = 0.25$ m to form a couple. The torque of this couple about any arbitrary point in the plane of the wheel is:
- (A) 5 N · m
 (B) 10 N · m
 (C) 2.5 N · m
 (D) Zero
- Q8.** A force $\vec{F} = 4\hat{i} + 5\hat{j}$ N acts on a particle located at point $P(2, 3)$ m. The torque of this force about the point $Q(1, 1)$ m is:
- (A) $+3\hat{k}$ N · m
 (B) $+13\hat{k}$ N · m
 (C) $-3\hat{k}$ N · m
 (D) $-13\hat{k}$ N · m
- Q9.** A uniform meter stick of mass $M = 0.6$ kg is balanced horizontally on a knife-edge placed at the 40 cm mark when a mass m is suspended at the 10 cm mark. The value of mass m is:
- (A) 0.3 kg
 (B) 0.4 kg
 (C) 0.2 kg
 (D) 0.15 kg
- Q10.** A particle of mass m is dropped from rest from a height h at horizontal distance d from the origin. The magnitude of torque of the gravitational force about the origin as a function of time t until it lands is:
- (A) $mgdt$
 (B) $\frac{1}{2}mgdt^2$
 (C) $mg(h - \frac{1}{2}gt^2)$
 (D) mgd
- Q11.** A solid circular disc of radius $R = 0.5$ m has three tangential forces acting along its rim: $F_1 = 10$ N (clockwise), $F_2 = 25$ N (counter-clockwise), and $F_3 = 5$ N (clockwise). The net torque acting on the disc about its central axle is:
- (A) 10.0 N · m clockwise

- (B) $5.0 \text{ N} \cdot \text{m}$ counter-clockwise
 (C) $2.5 \text{ N} \cdot \text{m}$ counter-clockwise
 (D) $7.5 \text{ N} \cdot \text{m}$ clockwise
- Q12.** The physical dimensions of torque are identical to the physical dimensions of:
- (A) Power
 (B) Work / Energy
 (C) Angular momentum
 (D) Linear momentum
- Q13.** A rectangular plate of sides $a = 0.4 \text{ m}$ and $b = 0.3 \text{ m}$ lies in the xy -plane with one corner at the origin. A force $\vec{F} = (10\hat{i} + 20\hat{j}) \text{ N}$ is applied at the opposite diagonal corner (a, b) . The torque of this force about the origin is:
- (A) $-5.0\hat{k} \text{ N} \cdot \text{m}$
 (B) $+11.0\hat{k} \text{ N} \cdot \text{m}$
 (C) $+8.0\hat{k} \text{ N} \cdot \text{m}$
 (D) $+5.0\hat{k} \text{ N} \cdot \text{m}$
- Q14.** A light horizontal rod AB of length L is pivoted at end A . Two forces F_1 and F_2 act perpendicularly to the rod at distances $L/3$ and L respectively from A in opposite senses. For rotational equilibrium ($\tau_{\text{net}} = 0$), the ratio F_1/F_2 must be:
- (A) 1 : 3
 (B) 2 : 3
 (C) 1 : 1
 (D) 3 : 1
- Q15.** If the net torque acting on a system of particles about a point is zero ($\vec{\tau}_{\text{ext}} = \vec{0}$), which of the following physical quantities must remain conserved about that point?
- (A) Linear momentum
 (B) Total mechanical energy
 (C) Angular momentum
 (D) Total kinetic energy
- Q16.** A uniform square lamina $ABCD$ of side length $a = 1 \text{ m}$ is acted upon by forces 10 N , 20 N , 30 N , and 40 N along the sides AB , BC , CD , and DA respectively in cyclic order. The magnitude of the net torque about the intersection of the diagonals is:
- (A) $50 \text{ N} \cdot \text{m}$
 (B) $100 \text{ N} \cdot \text{m}$
 (C) $25 \text{ N} \cdot \text{m}$
 (D) 0
- Q17.** A heavy door of width $W = 1.2 \text{ m}$ is pushed with a force $F = 40 \text{ N}$ applied at the outer handle perpendicular to the door surface. If the same torque is to be produced by pushing at a point 0.3 m from the hinges, the required perpendicular force is:
- (A) 160 N
 (B) 120 N
 (C) 80 N
 (D) 48 N
- Q18.** A force $\vec{F} = (2\hat{i} + 3\hat{j}) \text{ N}$ acts along the line $3x - 2y = 6$. The perpendicular lever arm distance $r_{\perp}$ of the line of action of the force from the origin is:
- (A) $\frac{6}{\sqrt{13}} \text{ m}$
 (B) $\frac{6}{5} \text{ m}$
 (C) $\sqrt{13} \text{ m}$
 (D) $\frac{3}{\sqrt{13}} \text{ m}$
- Q19.** Under which of the following conditions is the moment of a force (torque) about an axis strictly ZERO?
- (A) The line of action of the force intersects the axis of rotation
 (B) The line of action of the force is parallel to the axis of rotation
 (C) The magnitude of the force is zero
 (D) All of the above
- Q20.** A particle of mass m is executing uniform circular motion of radius R with constant angular speed ω in the xy -plane centered at the origin. The net torque acting on the particle about the origin is:
- (A) $\vec{0}$
 (B) $mR^2\omega^2\hat{k}$
 (C) $mR\omega^2\hat{k}$
 (D) $\frac{1}{2}mR^2\omega^2\hat{k}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-051)

Moment of a Force; Torque — Definition, Cross Product & Calculation

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	D	11	B	16	A
2	C	7	B	12	B	17	A
3	C	8	C	13	D	18	A
4	B	9	C	14	D	19	D
5	B	10	D	15	C	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (A)

The torque vector about the origin:

$$\begin{aligned}\vec{\tau} &= \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix} \\ &= \hat{i}(8 - (-9)) - \hat{j}(12 - 6) + \hat{k}(-9 - 4) \\ &= 17\hat{i} - 6\hat{j} - 13\hat{k} \text{ N} \cdot \text{m}\end{aligned}$$

Sol 2. Option (C)

At maximum height, horizontal coordinate is:

$$x = \frac{R}{2} = \frac{u^2 \sin 2\theta}{2g}$$

Downward gravitational force is $F_g = mg$.

Torque about the launch point:

$$\tau = F_g \cdot x = (mg) \left(\frac{u^2 \sin 2\theta}{2g} \right) = \frac{mu^2 \sin 2\theta}{2}$$

Sol 3. Option (C)

By vector definition, $\vec{\tau} = \vec{r} \times \vec{F}$ is mutually perpendicular to both the position vector $\vec{r}$ and the force vector $\vec{F}$ ($\vec{\tau} \cdot \vec{r} = 0$, $\vec{\tau} \cdot \vec{F} = 0$).

Sol 4. Option (B)

The torque about hinge A is:

$$\tau = rF \sin \phi = (2 \text{ m})(50 \text{ N}) \sin 30^\circ = 50 \text{ N} \cdot \text{m}$$

Sol 5. Option (B)

Position vector is $\vec{r} = x\hat{i} + 4\hat{j}$ m. Force is $\vec{F} = 6\hat{i}$ N. Torque about origin:

$$\vec{\tau} = (x\hat{i} + 4\hat{j}) \times (6\hat{i}) = -24\hat{k} \text{ N} \cdot \text{m}$$

Sol 6. Option (D)

For $\vec{\tau} = \vec{r} \times \vec{F} = \vec{0}$, the vectors $\vec{r}$ and $\vec{F}$ must be parallel/collinear:

$$\frac{2}{a} = \frac{-6}{3} = \frac{-12}{6} = -2 \implies a = -1$$

Sol 7. Option (B)

For a couple, the torque is independent of the reference point:

$$\tau = F(2R) = 20 \times (2 \times 0.25) = 10 \text{ N} \cdot \text{m}$$

Sol 8. Option (C)

Relative position vector:

$$\vec{r}_{P/Q} = \vec{r}_P - \vec{r}_Q = (2 - 1)\hat{i} + (3 - 1)\hat{j} = \hat{i} + 2\hat{j} \text{ m}$$

Torque about Q:

$$\vec{\tau} = (\hat{i} + 2\hat{j}) \times (4\hat{i} + 5\hat{j}) = (5 - 8)\hat{k} = -3\hat{k} \text{ N} \cdot \text{m}$$

Sol 9. Option (C)

Meter stick center of gravity is at 50 cm.

Fulcrum is at 40 cm (lever arm = 10 cm).

Mass m is at 10 cm (lever arm = 30 cm).

Torque equilibrium about the knife-edge:

$$mg(30) = Mg(10) \implies m = \frac{0.6}{3} = 0.2 \text{ kg}$$

Sol 10. Option (D)

The vertical line of action of gravity remains at constant perpendicular distance d from the origin throughout the fall:

$$\tau = F \cdot r_{\perp} = mgd$$

Sol 11. Option (B)

Taking counter-clockwise as positive:

$$\begin{aligned}\tau_{\text{net}} &= (F_2 - F_1 - F_3)R \\ &= (25 - 10 - 5)(0.5) \\ &= 5.0 \text{ N} \cdot \text{m} \text{ (counter-clockwise)}\end{aligned}$$

Sol 12. Option (B)

Dimensional formula of torque:

$$[\tau] = [r][F] = [L][M L T^{-2}] = [M L^2 T^{-2}]$$

This is identical to the dimensional formula of work and energy.

Sol 13. Option (D)

Position vector $\vec{r} = 0.4\hat{i} + 0.3\hat{j}$ m and $\vec{F} = 10\hat{i} + 20\hat{j}$ N:

$$\begin{aligned}\vec{\tau} &= (0.4\hat{i} + 0.3\hat{j}) \times (10\hat{i} + 20\hat{j}) \\ &= (0.4 \times 20 - 0.3 \times 10)\hat{k} = +5.0\hat{k} \text{ N} \cdot \text{m}\end{aligned}$$

Sol 14. Option (D)

For rotational equilibrium about pivot A:

$$\begin{aligned}F_1 \left(\frac{L}{3} \right) - F_2(L) &= 0 \\ \frac{F_1}{3} = F_2 &\implies \frac{F_1}{F_2} = \frac{3}{1}\end{aligned}$$

Sol 15. Option (C)

Since $\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$, zero net external torque ($\vec{\tau}_{\text{ext}} = \vec{0}$) ensures that total angular momentum $\vec{L}$ is conserved.

Sol 16. Option (A)

Perpendicular distance to each side is $r_{\perp} = a/2 = 0.5$ m.

All 4 forces produce torque in the same sense:

$$\begin{aligned}\tau_{\text{net}} &= (F_1 + F_2 + F_3 + F_4) \left(\frac{a}{2} \right) \\ &= (100)(0.5) = 50 \text{ N} \cdot \text{m}\end{aligned}$$

Sol 17. Option (A)

Torque: $\tau = F_1 W = 40 \times 1.2 = 48 \text{ N} \cdot \text{m}$.

At distance $d = 0.3$ m:

$$F_2 = \frac{\tau}{d} = \frac{48}{0.3} = 160 \text{ N}$$

Sol 18. Option (A)

Perpendicular distance from origin to line $3x - 2y - 6 = 0$:

$$r_{\perp} = \frac{|-6|}{\sqrt{3^2 + (-2)^2}} = \frac{6}{\sqrt{13}} \text{ m}$$

Sol 19. Option (D)

Torque about an axis is zero if $F = 0$, or if the line of action intersects the axis ($r_{\perp} = 0$), or if the force is parallel to the axis ($\vec{F} \parallel \text{axis}$). Thus, all options are correct.

Sol 20. Option (A)

In uniform circular motion, centripetal force is strictly radial ($\vec{F}_c \parallel -\vec{r}$):

$$\vec{\tau} = \vec{r} \times \vec{F}_c = \vec{0}$$